Brain Tumor Survival Prediction using Deep Neural Networks

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1 Introduction

In this report, we will take you through our project titled "MRI Brain Tumor Survival Prediction" available on Kaggle. Our main objective was to develop a machine learning model that could accurately predict the survival outcome of brain tumor patients using MRI scans and other clinical features. The dataset we worked with contained MRI images and related information, such as age, tumor size, and survival status. Our goal was to leverage this data to build a predictive model that could assist in prognosis and treatment planning.

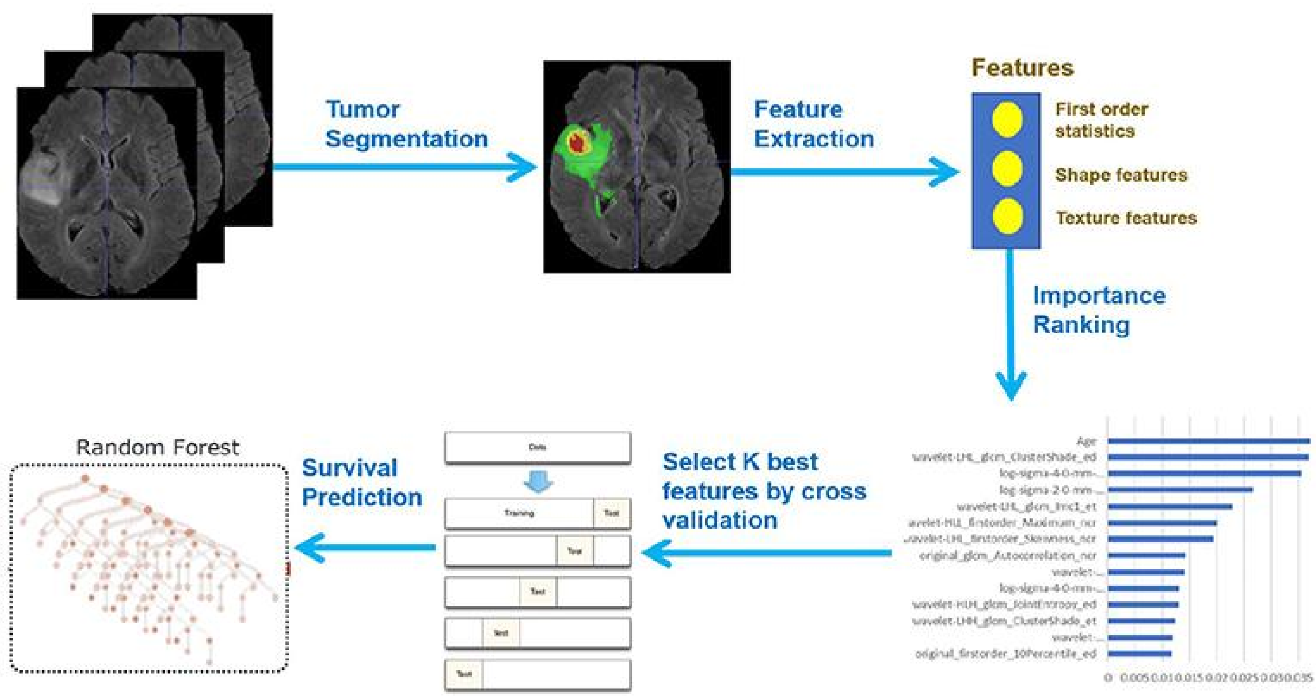


Figure 1: Framework overview

2 Dataset

BraTS has always been focusing on the evaluation of state-of-the-art methods for the segmentation of brain tumors in multimodal magnetic resonance imaging (MRI) scans. BraTS 2020 utilizes multi-institutional pre-operative MRI scans and primarily focuses on the segmentation (Task 1) of intrinsically heterogeneous (in appearance, shape, and histology) brain tumors, namely gliomas. Furthemore, to pinpoint the clinical relevance of this segmentation task, BraTS'20 also focuses on the prediction of patient overall survival (Task 2), and the distinction between pseudoprogression and true tumor recurrence (Task 3), via integrative analyses of radiomic features and machine learning algorithms. Finally, BraTS'20 intends to evaluate the algorithmic uncertainty in tumor segmentation (Task 4).

Imaging Data Description

All BraTS multimodal scans are available as NIfTI files (.nii.gz) and describe a) native (T1) and b) post-contrast T1-weighted (T1Gd), c) T2-weighted (T2), and d) T2 Fluid Attenuated Inversion Recovery (T2-FLAIR) volumes, and were acquired with different clinical protocols and various scanners from multiple (n=19) institutions, mentioned as data contributors here.

All the imaging datasets have been segmented manually, by one to four raters, following the same annotation protocol, and their annotations were approved by experienced neuro-radiologists. Annotations comprise the GD-enhancing tumor (ET — label 4), the peritumoral edema (ED — label 2), and the necrotic and non-enhancing tumor core (NCR/NET — label 1), as described both in the BraTS 2012-2013 TMI paper and in the latest BraTS summarizing paper. The provided data are distributed after their pre-processing, i.e., co-registered to the same anatomical template, interpolated to the same resolution (1 mm^3) and skull-stripped.

3 Project Overview

Our project began with thorough data preprocessing and exploration. We started by standardizing the MRI images to eliminate bias and ensure consistency. This involved applying normalization techniques to adjust the intensity values across the images. By standardizing the images, we aimed to create a level playing field for our model to learn from. Following preprocessing, we delved into exploratory data analysis to gain insights into the dataset. Visualizations were instrumental in understanding the distribution of tumor sizes, age demographics, and survival outcomes. These explorations paved the way for our subsequent modeling steps.

4 Architecture

The cornerstone of our project was a Convolutional Neural Network (CNN) model. CNNs are powerful deep learning architectures specifically designed for image analysis tasks. The architecture consists of multiple layers, including convolutional layers, pooling layers, and fully connected layers. The convolutional layers are responsible for extracting relevant features from the input MRI images. By applying a series of convolutional filters, the CNN can learn different patterns and textures present in the images. The pooling layers reduce the spatial dimensions of the extracted features, enabling the model to capture higher-level representations. Finally, the fully connected layers combine the extracted features and make predictions based on the learned information.Our architecture comprises of convolutional blocks that extract survival related features from the MRI images concatenated with their corresponding segmented image(as a channel map), concatenates age of the patient with these feature maps and finally Fully Connected layers are applied. The final prediction has been distributed into 2 models : A classification model that classifies the patient to lie inone of the following 3 categories :-

- Survival days lying in range 0 - 300
- Survival days lying in range 300 - 450
- Survival days lying in range 450+

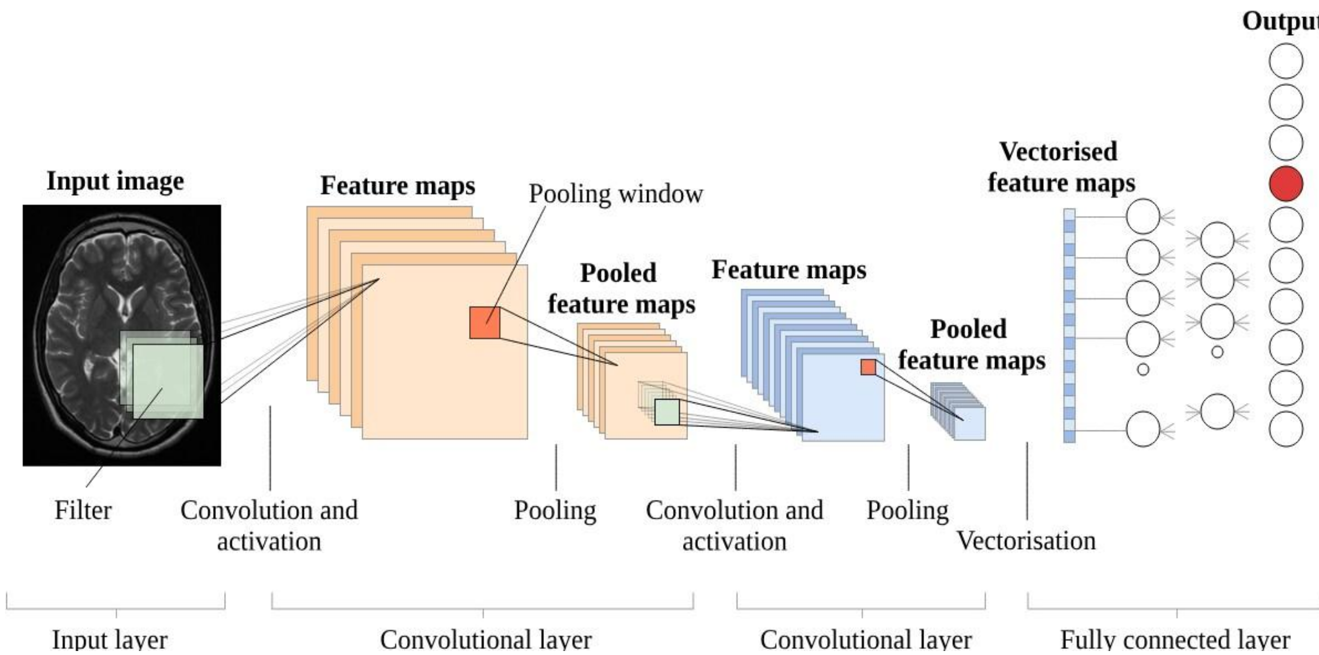


Figure 2: CNN Architecture

4.1. Integration of Clinical Features:

To enhance the predictive capabilities of our CNN model, we incorporated additional clinical features, such as age and tumor size, into its architecture. By integrating patient-specific information, we aimed to capture the influence of these factors on survival outcomes. This integration involved concatenating the clinical features with the output of the convolutional layers before feeding them into the fully connected layers. By doing so, the model could learn the relationships between the MRI images, clinical features, and survival outcomes, enabling more accurate predictions.

4.2. Random Forest, SVM, KNN, and Grid Search:

In addition to the CNN model, we explored and integrated other machine learning algorithms . We implemented Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) algorithms. Random Forest is an ensemble learning method that combines multiple decision trees to make predictions. SVM is a powerful algorithm for classification and regression tasks, utilizing a hyperplane to separate data points into different classes. KNN is a non-parametric algorithm that classifies data points based on their proximity to neighboring points.

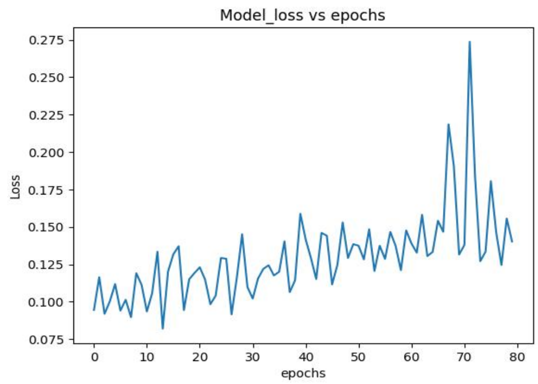
To optimize the hyperparameters of these models, we utilized Grid Search, a technique that exhaustively searches for the best combination of hyperparameter values. Grid Search involves defining a set of possible hyperparameter values and evaluating the model's performance for each combination. By systematically testing different parameter combinations, we aimed to identify the optimal configuration for each algorithm, maximizing their predictive capabilities.

5 Model Training and Evaluation

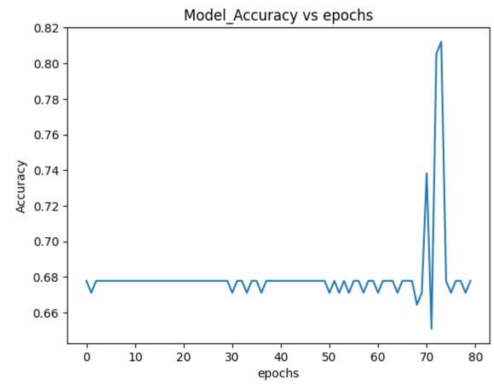
5.1 Training Details

We have total data for 118 patients. We split it into 94 and 24(0.2%) for training and testing respectively. The classification model is trained on 94 patients with Adam optimiser, Categorical Cross Entropy as loss function and accuracy as metric. With learning rate set to 1e-2, we train for 80 epochs with batch size 64.

We split the dataset into training and testing sets to train and evaluate our models.The training set has 94 samples and the testing set has 24 samples.For the CNN model, we used the training set to optimize its parameters and minimize the loss function. We monitored the model's performance using evaluation metrics such as accuracy, precision, recall, and F1 score. These metrics provided insights into the model's ability to correctly classify survival outcomes. We also compared the model's predictions with the ground truth survival outcomes to assess its accuracy and effectiveness.



(a) Loss vs epochs



(b) Accuracy vs epochs

For the Random Forest, SVM, and KNN models, we trained them on the training set and evaluated their performance using similar metrics. The hyperparameters of these models were tuned using Grid Search, which involved iterating through different parameter combinations and selecting the configuration that yielded the best performance.

5.2 Evaluation/Prediction and Results

Our model produced an accuracy of **0.4583** on the test set of 24 samples. Our project's results demonstrated the potential of utilizing MRI images and clinical features for survival prediction in brain tumor patients. The CNN model, with the integration of clinical features, showed promising predictive capabilities. However, the performance of the models varied across different algorithms. The Random Forest, SVM, and KNN models also provided reasonable predictions, although their accuracy might be influenced by the specific characteristics of the dataset and patient population.

6 Conclusion

Our project on "MRI Brain Tumor Survival Prediction" has showcased the application of machine learning techniques in predicting the survival of brain tumor patients using MRI scans and clinical features. By leveraging a CNN model with integrated clinical information, we have demonstrated the potential to improve prognostic predictions and aid in treatment planning. Furthermore, the exploration of Random Forest, SVM, and KNN algorithms, along with hyperparameter tuning using Grid Search, has provided valuable insights into their performance for this prediction task.

It is important to note that further research and validation are required to establish the robustness and reliability of the models on diverse datasets and in real-world clinical scenarios. Nonetheless, our project contributes to the growing field of medical imaging analysis and highlights the significance of combining multiple data modalities for enhanced patient outcome prediction in healthcare. We try to build a deep learning architecture that makes use of our segmentation model.

References

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[4] Fabian Isensee, Philipp Kickingereder, Wolfgang Wick, Martin Bendszus, andKlaus H. Maier-Hein. Brain tumor segmentation and radiomics survival predic tion: Contribution to the BRATS 2017 challenge. CoRR, abs/1802.10508, 2018.